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 Studierichting: Informatica
 Oefeningenreeks 3A nr. 40.4

$$f(x) = \sqrt{x^3 + 2x^2 + 2 - \frac{1}{x} - \frac{3}{x^2}}$$

$$f'(x) = \frac{3x^2 + 4x + \frac{1}{x^2} + \frac{6}{x^3}}{2\sqrt{x^3 + 2x^2 + 2 - \frac{1}{x} - \frac{3}{x^2}}}$$

tussenstap 1:

$$\frac{3x^2 \cdot x^3}{x^3} + \frac{4x \cdot x^3}{x^3} + \frac{x}{x^3} + \frac{6}{x^3}$$

$$= \frac{3x^5 + 4x^4 + x + 6}{x^3}$$

tussenstap 2:

$$\sqrt{x^3 + 2x^2 + 2 - \frac{1}{x} - \frac{3}{x^2}}$$

$$= \sqrt{\frac{x^3 \cdot x^2}{x^2} + \frac{2x^2 \cdot x^2}{x^2} + \frac{2x^2}{x^2} - \frac{x}{x^2} - \frac{3}{x^2}}$$

$$= \sqrt{\frac{x^5 + 2x^4 + 2x^2 - x - 3}{x^2}}$$

$$= \frac{\sqrt{x^5 + 2x^4 + 2x^2 - x - 3}}{\sqrt{x^2}}$$

$$= \frac{\sqrt{x^5 + 2x^4 + 2x^2 - x - 3}}{|x|}$$

Pas deze twee tussenstappen toe:

$$\frac{3x^5 + 4x^4 + x + 6}{x^3}$$

$$\Rightarrow \frac{\frac{3x^5 + 4x^4 + x + 6}{x^3}}{2 \frac{\sqrt{x^5 + 2x^4 + 2x^2 - x - 3}}{|x|}}$$

$$= \frac{3x^5 + 4x^4 + x + 6}{x^3} \cdot \frac{x}{2\sqrt{x^5 + 2x^4 + 2x^2 - x - 3}}$$

$$= \frac{3x^5 + 4x^4 + x + 6}{2x \cdot |x| \cdot \sqrt{x^5 + 2x^4 + 2x^2 - x - 3}}$$

References